Summary of Lagrangian Duality (Part I)

Large Scale Optimization

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1. Basic Definitions and Properties

Basic Definitions

Reference Problem and Lagrangian

$$\begin{array}{ll} \text{Min}_x & f(x) \\ \text{s.t. :} & h(x) = 0 \mid \lambda \in \mathbb{R}^p \\ & g(x) \geq 0 \mid \mu \in \mathbb{R}^q_+ \\ & x \in X \end{array}$$

Associated Lagrangian function:

$$\mathcal{L}(x, \lambda, \mu) = f(x) - \lambda^\top h(x) - \mu^\top g(x)$$

Dual Lagrangian Function; Dual Lagrangian Problem

$$\begin{array}{lll} w(\lambda, \mu) \triangleq \text{Min}_x & \mathcal{L}(x, \lambda, \mu) & \\ (\text{LD}_0) & x \in X & \text{Max}_x \quad w(\lambda, \mu) \\ & \downarrow & (\text{LD}) \quad (\lambda, \mu) \in D \\ (\text{S}_{LD}(\lambda, \mu) \text{ the} & \text{solution set}) & \end{array}$$

$$D \triangleq \{ (\lambda, \mu) \in \mathbb{R}^p \times \mathbb{R}^q_+ \mid w(\lambda, \mu) > -\infty \}$$

Usually, $D \equiv \mathbb{R}^p \times \mathbb{R}^q_+$

Properties of $w(\lambda, \mu)$

Weak Duality Theorem

Let $(\lambda, \mu) \in D$, x feasible primal (of problem (P)). Then $w(\lambda, \mu) \leq f(x)$

- gap$(x, \lambda, \mu) \stackrel{\Delta}{=} f(x) - w(\lambda, \mu) =$
$f(x) - f(\bar{x}) + \lambda^\top h(\bar{x}) + \mu^\top g(\bar{x}) \geq 0,\ \bar{x} \in \mathsf{S}_{LD}(\lambda, \mu)$.
- If (λ^*, μ^*) solves (LD) and $\bar{x} \in \mathsf{S}_{LD}(\lambda^*, \mu^*)$, then
gap$(\bar{x}, \lambda^*, \mu^*) = \lambda^{*\top} h(\bar{x}) + \mu^{*\top} g(\bar{x})$
- If gap$(\bar{x}, \lambda^*, \mu^*) = \lambda^{*\top} h(\bar{x}) + \mu^{*\top} g(\bar{x}) = 0$, then $\bar{x}$ solves (P)

Other properties. (See Bazaraa's textbook, Chapter 6)

- $w(\lambda, \mu)$ is concave on D
- The subgradient set of $w(\lambda, \mu)$ at a point $(\bar{\lambda}, \bar{\mu})$ is given by:

$$\partial w(\lambda, \mu) = \text{Hull}\left(\left\{ \begin{pmatrix} -h(x^*) \\ -g(x^*) \end{pmatrix} \;\middle|\; x^* \in \mathsf{S}_{LD}(\lambda, \mu) \right\}\right)$$

2. Types of Problems

Types of Problems

- **Convex Problems**. (including LP's) $f(x)$ convex, h affine, $g(x)$ locally convex at the solution (active constraints). **Have null duality gap**.
- Problems with $w(\lambda, \mu)$ **differentiable** also **have null duality gap**.
- **IP's, MILP's may have non-null duality gap**. Additional Lagrangian heuristics are needed to obtain a good solution.

Integer Programming Problems

Reference Problem and Lagrangian Dual

$$\begin{array}{lll} z^* = \mathsf{Min}_x & c^\top x & \\ \quad\quad \text{s.t.}: & Ax = b \mid \lambda & \\ & Bx = d & \\ \text{(P)} & x \in \mathbb{Z}^n & \end{array}$$

$$\begin{array}{lll} z^*_{\mathsf{LR}} = \mathsf{Min}_x & c^\top x & \\ \quad\quad \text{s.t.}: & Ax = b \mid \lambda & \\ \text{(LR)} & x \in X_{\mathsf{LR}} & \end{array}$$

$$\begin{array}{lll} w(\lambda) = \mathsf{Min}_x & c^\top x - \lambda^\top (Ax - b) & \\ \text{(LD}_0\text{)} & x \in X & \end{array}$$

$$\begin{array}{lll} z^*_{\mathsf{LD}} = \mathsf{Max}_\lambda & w(\lambda) & \\ \text{(LD)} & \lambda \in D & \end{array}$$

$$X = \{\, x \in \mathbb{Z}^n \mid Bx = d \,\}$$
$$X_{\mathsf{LR}} = \{\, x \in \mathbb{R}^n \mid Bx = d \,\}$$

Theorem

$$\begin{array}{llll} z^*_{\mathsf{LD}} = \mathsf{Max}_\lambda & w(\lambda) & = \mathsf{Min}_x & c^\top x \\ \text{(LD)} & \lambda \in D & \text{s.t. :} & Ax = b \\ & & & x \in \mathsf{Hull}(X) \end{array}$$

Consequences

$$z^*_{\mathsf{LR}} \overset{(a)}{\leq} z^*_{\mathsf{LD}} \overset{(b)}{\leq} z^*$$

- if $\mathsf{Hull}(X) = X_{\mathsf{RL}}$, then (a) becomes $=$
- if $\mathsf{Hull}(\{x \in X | Ax = b\}) = \mathsf{Hull}(X) \cap \{x \in \mathbb{R}^n | Ax = b\}$, then (b) becomes $=$
- In general $\mathsf{Hull}(X) \subset X_{\mathsf{LR}}$ and $z^*_{\mathsf{LR}} < z^*_{\mathsf{LD}} \leq z^*$.

  If $X = \{x \in \mathbb{Z}^2_+ | 2x_1 + 2x_2 \leq 5\}$, then
  $\mathsf{Hull}(X) = \{x \in \mathbb{R}^2_+ | x_1 + x_2 \leq 2\}$ and
  $X_{\mathsf{LR}} = \{x \in \mathbb{R}^2_+ | 2x_1 + 2x_2 \leq 5\}$

3. Solving (LD)

1st Order Optimality Conditions. (1st OOC)

Assume that for the problem (P), problem (LD) reduces to:

$$\begin{array}{ll} \text{Max}_x & w(\lambda,\mu) \\ \text{(LD)} & \mu \geq 0 \quad | \theta \geq 0 \end{array}$$

1st OOC. Assume w differentiable

$$\begin{aligned} &\nabla_\lambda w(\lambda^*,\mu^*) = 0 \\ &-\nabla_\mu w(\lambda^*,\mu^*) = \theta \geq 0,\ \theta^\top \mu^* = 0,\ \mu^* \geq 0 \end{aligned}$$

1st OOC. Assume w non-differentiable

$$\begin{aligned} &\exists r \in \partial w(\lambda^*,\mu^*) \text{ such that :} \\ &\ r_\lambda = 0 \\ &-r_\mu = \theta \geq 0,\ \theta^\top \mu^* = 0,\ \mu^* \geq 0 \end{aligned}$$

In variational form:

$$\partial w(\lambda^*,\mu^*) \in \mathsf{N}_{\mathbb{R}^p \times \mathbb{R}^q_+}(\lambda^*,\mu^*) \equiv \left\{ \begin{array}{l} r_\lambda = 0 \\ r_\mu \leq 0,\ r_\mu^\top \mu^* = 0 \end{array} \right.$$

Dual Search Methods

Methods for solving the Lagrangian Dual Problem (LD)

$$\begin{array}{ll} \text{Max}_x & w(\lambda, \mu) \\ \text{(LD)} & (\lambda, \mu) \in D \end{array}$$

- Subgradient method

  (see `subgradient-algorithm.pdf`)
- Outer approximation method (Dantzig's cutting plane)

  (see `DANTZIG-CUTTING-PLANE1-AP8.pdf`)
- Bundle Methods (Lemarechal)
- Non-differentiable optimization methods

  (see `wolfe12.pdf`, `wolfe13.pdf`)